

En: 14
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Write a code using RAW sockets to
implement packet sniffing.

```
from scapy.all import sniff
from scapy.layers.inet import IP, TCP, UDP, ICMP

def packet_callback(packet, ):
    if IP in packet:
        ip_layer = packet[IP]
        protocol = ip_layer.proto
        src_ip = ip_layer.src
        dest_ip = ip_layer.dst

        # Determine the protocol.
        protocol_name = ""
        if protocol == 1:
            protocol_name = "ICMP"
        elif protocol == 6:
            protocol_name = "TCP"
        elif protocol == 17:
            protocol_name = "UDP"
        else:
            protocol_name = "Unknown Protocol"

        print(f"Protocol: {protocol_name}")
        print(f"Source IP: {src_ip}")
        print(f"Destination IP: {dest_ip}")
        print("-" * 50)

sniff (iface="Wi-Fi", prn=packet_callback,
      filter="ip", store=0)
```

Op.

Destination IP: 224.0.0.251

Protocol: TCP

Source IP: 172.16.11.15

Destination IP: 172.16.8.1

Protocol: UDP

Source IP: 172.16.11.136

Destination IP: 239.255.255.259.

~~Result~~

Thus, the Python program was successfully executed to check the ^{socket} programming.

Messages were successfully sent and received between the client and server.